



Gosford High School

Year 11

2008

Preliminary Higher School Certificate

Mathematics Extension 1 Assessment Task 4

Time Allowed – 1 hour 30 mins + 5 mins reading

Remember to start each new question on a new page

Students must answer questions using a blue/black pen, and a sharpened B or HB pencil for graphs.

Approved scientific calculators may be used

Students need to be aware that

- * 'bald' answers may not gain full marks.
- * untidy and/or poorly organised solutions may not gain full marks.

Mathematics Extension 1 – Preliminary Higher School Certificate – 2008

Question 1 (12 Marks) Review

Marks

- a) Solve $\frac{2}{x-1} \leq 1$ **2**
- b) A is the point (1, 2) and B is the point (-1, -4). Find the coordinates of the point P which divides AB externally in the ratio 1 : 2. **3**
- c) Find the obtuse angle between the lines
 $3x - y = 1$ and $x + 2y = 4$. (Answer to nearest minute.) **3**
- d) Solve $\frac{x^2-6}{x} \leq 1$ **4**

Question 2 (15 Marks) Calculus (Begin a new sheet of paper)

- a) Differentiate:
- (i) $\sqrt{2}x - \sqrt{x} + \frac{x^2}{\sqrt{x}} - \frac{1}{2x}$ **4**
- (ii) $x(3x-2)^5$ and simplify your answer **3**
- (iii) $\frac{2x+1}{x-3}$ and simplify your answer **2**
- b) Differentiate from first principles: $f(x) = 2x - 3x^2$ **3**
- c) Find the equation of the normal to the curve $y = 3x^2 - 4x + 7$ at the point where $x=1$. **3**

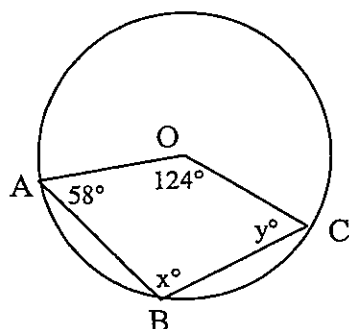
Question 3 (14 Marks) Quadratic Function & Polynomials (Begin a new sheet of paper) Marks

- a) Find the values of m for which the expression $x^2 + (m-2)x + 4$ is positive definite. 3
- b) Find the values of a , b , and c if $3x^2 - 5x + 2 \equiv a(x+1)^2 + b(x+1) + c$ 3
- c) (i) Show that $x+1$ is a factor of $P(x) = -x^3 + 3x + 2$ 1
- (ii) Fully factorise $P(x)$ 4
- (iii) Hence sketch the graph of $y = P(x)$ 3

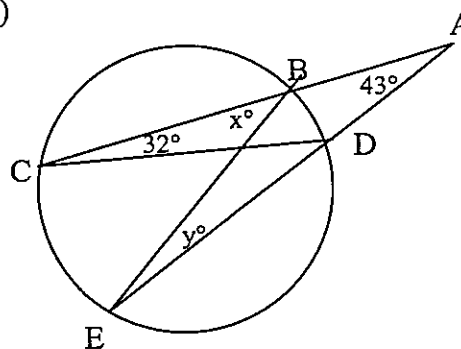
Question 4 (12 Marks) Circle Geometry (Begin a new sheet of paper)

- a) Find the value of the pronumerals x and y , giving all reasons

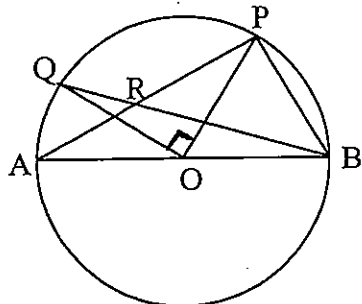
- (i) O is the centre 3



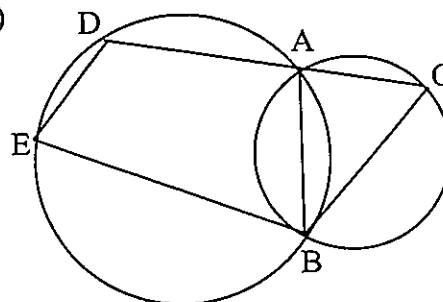
- (ii) 3



- b) (i) 3



- (ii) 3



O is the centre.

$$\angle POQ = 90^\circ$$

Find the size of $\angle PBQ$

Hence, prove that $PB = PR$

EB is a tangent to circle ABC

Prove that $BC \parallel ED$

Question 5 (12 Marks) (Begin a new sheet of paper)**Marks**

- a) (i) Write $\sqrt{3} \sin \theta + \cos \theta$ in the form $R \cos(\theta - \alpha)$ 3
- (ii) Solve $\sqrt{3} \sin \theta + \cos \theta = 1$ for $-180^\circ \leq \theta \leq 360^\circ$ 3
- b) The roots of $x^3 - x^2 - 5x + 2 = 0$ are α , β and γ .
- (i) Write down the values of $\alpha\beta\gamma$ and $\alpha\beta + \beta\gamma + \gamma\alpha$ 2
- (ii) Show that $\beta + \gamma = 1 - \alpha$ 1
- (iii) **Hence** evaluate $\frac{\alpha + \beta}{\gamma} + \frac{\beta + \gamma}{\alpha} + \frac{\gamma + \alpha}{\beta}$ 3

Question 6 (13 Marks) Extension (Begin a new sheet of paper)

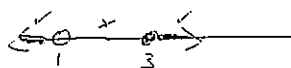
- a) If α and β are the roots of $2x^2 + 5x - 3 = 0$, form the equation with roots α^2 and β^2 3
- b) Express the derivative of $y = \frac{2x+1}{\sqrt{3x-2}}$ in simplest form. 3
- c) The point $P(x, y)$ divides the join of A and B in the ratio $k : 1$
If A is the point $(1, 2)$ and B is the point $(-1, -4)$,
- (i) Write expressions for x and y in terms of k . 2
- (ii) **Hence** find the value of k , such that the line $3x - 4y - 5 = 0$ divides AB in the ratio $k : 1$ 2
- d) Find as a relation between k , l , and m , the condition for the quadratic equation in x
$$(k^2 + l^2)x^2 + 2l(k+m)x + (l^2 + m^2) = 0$$
 3
to have real roots. Simplify your answer as far as possible.

21) $\frac{2}{x-1} \leq 1$ CP at $x=1, x \neq 1$

Make equation to find other CP's

$2 = x-1$

$3 = x$



$x \geq 3$ and $x < 1$

(b) $(1, 2) \rightarrow (-1, -4)$

$(-1, 2)$

$\left(\frac{(-1)(-1) + (2)(1)}{1}, \frac{(-1)(4) + (2)(2)}{1} \right)$

$(3, 8)$

(3)

(c) $y = 3x-1$ $2y = -x+4$
 $m_1 = 3$ $m_2 = -\frac{1}{2}$

$\tan \theta = \left| \frac{3 + \frac{1}{2}}{1 - (3)(-\frac{1}{2})} \right|$

$\tan \theta = 1.7$

$\theta = 81^\circ 52'$ but obtuse

$\therefore \theta = 98^\circ 8'$ (3)

(d) Solve $\frac{x^2-6}{x} \leq 1$ CP at $x=0, x \neq 0$

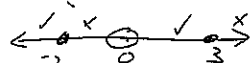
Make equation to find other CP's

$x^2-6 = x$

$x^2-x-6=0$

$(x-3)(x+2)=0$

CP's at $x=3, -2$



$0 < x \leq 3$ and $x \leq -2$

(4)

QUESTION 3

$\Delta < 0$

$b^2 - 4ac < 0$

$m-6)(m+2) < 0$

$-2 < m < 6$

$3x^2 - 5x + 2 \equiv a(x^2 + 2x + 1) + b(x+1) + c$

$\equiv ax^2 + 2ax + a + bx + b + c$

$a = 3$

$2a + b = -5$

$b = -11$

$a + b + c = 2$

$3 - 11 + c = 2$

$c = 10$

or sub $x = -1$ gives $c = 10$

$P(-1) = -(-1)^2 + 3(-1) + 2$

$= 0$ $\therefore$ factor

i) $x+1$ $\frac{-x^2+x+2}{x+1}$

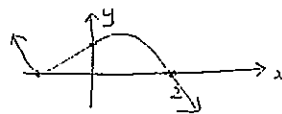
$(x+1)(-x^2+x+2)$

$-(x+1)(x^2-x-2)$

$-(x+1)(x-2)(x+1)$

$-(x+1)^2(x-2)$

iv)



cuts $x = 2$

turns $x = -1$

start under

y intercept 2

no need label

QUESTION 4

i) $\alpha = \frac{1}{2} \times 236^\circ$ $\angle$ centre = 2 $\times$ angle circle

$= 118^\circ$

$y = 60^\circ$ $\angle$ sum quad

ii) $y = 33^\circ$ $\angle$ on same arc

$\angle CDE = 75^\circ$ $\angle$ exterior angle

$\angle x = 75^\circ$ $\angle$ on same arc

b) $\angle APB = 90^\circ$ ($\angle$ in semi circle)

$\angle PBQ = 45^\circ$ ($\angle$ centre 2 $\times$ angle circum)

$\angle PRB = 45^\circ$ ($\angle$ sum $\angle$)

$\therefore$ PRB isosceles $\triangle$ 2 base $\angle$ s equal

ii) $\angle EBA = \angle BCA$ ($\angle$ angle in alternate seg)

$\angle EDA = 180 - x$ opp $\angle$ in cyclic quad

$\angle EDA + \angle DCB = 180^\circ$ (co-interior $\angle$ s)

$\therefore$ BB || DE

$$\cos \theta + \sqrt{3} \sin \theta$$

$$\cos(\theta - \alpha) = R \cos \theta \cos \alpha + R \sin \theta \sin \alpha$$

$$\therefore R \cos \alpha = 1$$

$$R \sin \alpha = \sqrt{3}$$

$$\tan \alpha = \sqrt{3}$$

$$\alpha = 60^\circ$$

$$R = \sqrt{3+1}$$

$$= 2$$

$$2 \cos(\theta - 60^\circ) = 2 \cos(\theta - 60^\circ)$$

$$2 \cos(\theta - 60^\circ) = 1$$

$$\cos(\theta - 60^\circ) = \frac{1}{2}$$

$$\theta - 60^\circ = 60^\circ, 300^\circ \text{ or } -60^\circ$$

$$\theta = 120^\circ, 360^\circ \text{ or } 0^\circ$$

$$1) \alpha\beta\gamma = -2$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = -5$$

$$2) \alpha + \beta + \gamma = -\frac{b}{a}$$

$$= 1$$

$$\therefore \beta + \gamma = 1 - \alpha$$

$$\frac{\alpha + \beta}{\gamma} + \frac{\beta + \gamma}{\alpha} + \frac{\gamma + \alpha}{\beta}$$

$$\frac{1 - \gamma}{\gamma} + \frac{1 - \alpha}{\alpha} + \frac{1 - \beta}{\beta}$$

$$\frac{\alpha\beta - \alpha\gamma + \beta\gamma - \alpha\beta\gamma + \alpha\gamma - \alpha\beta\gamma}{\alpha\beta\gamma}$$

$$\frac{\alpha\beta + \beta\gamma + \gamma\alpha - 3\alpha\beta\gamma}{\alpha\beta\gamma}$$

$$\frac{-5+6}{-2}$$

$$-\frac{1}{2}$$

26 (c) (i)

$$\begin{matrix} A & B \\ (1, 2) & (-1, -4) \\ K: 1 \end{matrix}$$

$$\frac{(K)(-1) + (1)(1)}{K+1}, \frac{(K)(-4) + (1)(2)}{K+1}$$

$$\left(\frac{1-K}{K+1}, \frac{2-4K}{K+1} \right) \quad (2)$$

$$(ii) \frac{3(1-K)}{K+1} - \frac{2-4K}{K+1} - \frac{5(K+1)}{(K+1)} = 0$$

$$3 - 3K - 8 + 16K - 5K - 5 = 0$$

$$-10 + 8K = 0$$

$$8K = 10$$

$$K = \frac{10}{8}$$

$$K = \frac{5}{4}$$

(2)

1) Real roots if $b^2 - 4ac \geq 0$

$$[2L(k+m)]^2 - 4(k^2 + L^2)(L^2 + m^2) \geq 0$$

$$4L^2(k^2 + 2km + m^2) - 4(k^2L^2 + k^2m^2 + L^4 + L^2m^2) \geq 0$$

$$4k^2L^2 + 8kL^2m + 4L^2m^2 - 4k^2L^2 - 4k^2m^2 - 4L^4 - 4L^2m^2 \geq 0$$

$$-4L^4 + 8kL^2m - 4k^2m^2 \geq 0$$

$$-4(L^4 - 2kL^2m + k^2m^2) \geq 0$$

$$L^4 - 2kL^2m + k^2m^2 \leq 0$$

$$L^2 \quad -km$$

$$L^2 \quad -km$$

$$(L^2 - km)^2 \leq 0$$

As left hand side must be positive

or zero

$$\text{then } L^2 = km$$

$$(a) \quad 2 = \frac{1}{2} - \frac{1}{2}$$

$$(\alpha + \beta) = -\frac{5}{2} \quad \alpha\beta = -\frac{3}{2}$$

$$\begin{aligned} \alpha^2 + \beta^2 &= (\alpha + \beta)^2 - 2\alpha\beta & (\alpha\beta)^2 &= \frac{9}{4} \\ &= \frac{25}{4} + \frac{6}{2} \\ &= \frac{37}{4} \end{aligned}$$

$$\therefore x^2 - (\alpha^2 + \beta^2)x + (\alpha\beta)^2 = x^2 - \frac{37}{4}x + \frac{9}{4}$$

$$= 4x^2 - 37x + 9$$

$\therefore$ required equation is

$$4x^2 - 37x + 9 = 0$$

$$(b) \quad y = \frac{2x+1}{(3x-2)^{\frac{1}{2}}}$$

$$y' = \frac{(3x-2)^{\frac{1}{2}} \cdot 2 - [(2x+1) \cdot \frac{1}{2} (3x-2)^{-\frac{1}{2}} \cdot 3]}{(3x-2)^{\frac{3}{2}}}$$

$$= \frac{(3x-2)^{\frac{1}{2}} [2(3x-2) - \frac{3}{2}(2x+1)]}{(3x-2)^{\frac{3}{2}}}$$

$$= \frac{4(3x-2) - 3(2x+1)}{(3x-2)^{\frac{3}{2}}}$$

$$= \frac{12x - 8 - 6x - 3}{2\sqrt{(3x-2)^3}}$$

$$= \frac{6x - 11}{2\sqrt{(3x-2)^3}}$$

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